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Physics
Assignment - 2
Vectors

classmate

Date _____

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1. Vectors & Scalars

A scalar is a quantity that has magnitude

A vector is a quantity that has magnitude and direction

Temperature - Scalar

Acceleration - Vector

Height - Scalar

Distance - Scalar

Velocity - Vector

Weight - Scalar

Mass - Scalar

Force - Vector

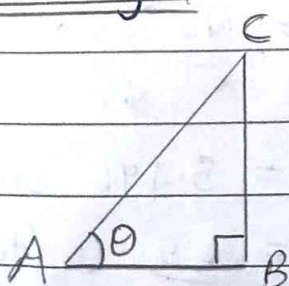
Speed - Scalar

Time - Scalar

Scalar - Temperature, Height, Mass, Speed, distance, weight, time

Vector - velocity, acceleration, force

2. Trigonometry

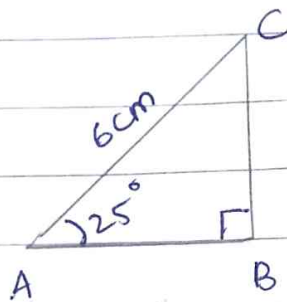


Formulas associated with $\triangle ABC$:

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{BC}{AC}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{AB}{AC}$$

$$\frac{BC}{AB} \Leftrightarrow \frac{\text{opposite}}{\text{adjacent}} \Leftrightarrow \tan \theta \Rightarrow \frac{\sin \theta}{\cos \theta} = \frac{\frac{BC}{AC}}{\frac{AB}{AC}} = \frac{BC}{AB}$$



$AC = 6 \text{ cm}$ $AB = ?$ $BC = ?$

$$\sin 25 = \frac{BC}{6} \quad (= (\sin \theta = \frac{\text{op}}{\text{hy}} = \frac{BC}{AC}))$$

$$BC = (\sin 25) \times 6$$

$$= 0.42 \times 6$$

$$= 2.52 \text{ cm}$$

Ans: $BC = 2.52 \text{ cm}$

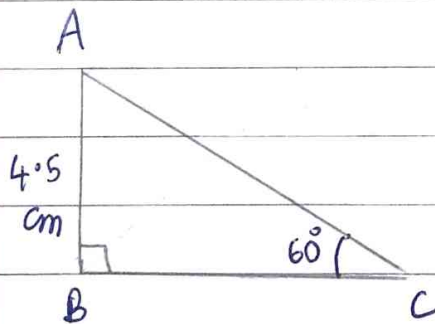
$AB = 5.43 \text{ cm}$

$$\cos 25 = \frac{AB}{6} \quad (= (\cos \theta = \frac{\text{ad}}{\text{hy}} = \frac{AB}{AC}))$$

$$AB = (\cos 25) \times 6$$

$$= 0.906 \times 6$$

$$= 5.43 \text{ cm}$$



$AB = 4.5 \text{ cm}$ $AC = ?$ $BC = ?$

$$\sin 60 = \frac{4.5}{AC} \quad (= (\sin \theta = \frac{\text{op}}{\text{hy}} = \frac{AB}{AC}))$$

$$AC = \frac{4.5}{\sin 60} = \frac{4.5}{\frac{\sqrt{3}}{2}} = \frac{9}{\sqrt{3}}$$

Ans: $AC = 5.196$, $BC = 2.598$

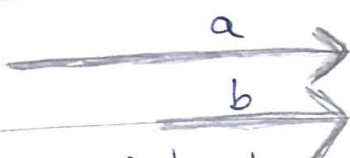
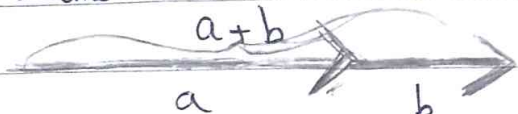
$$= \frac{(\sqrt{3} \cdot \sqrt{3}) \times 3}{\sqrt{3}} = 3\sqrt{3}$$

$$= 5.196 \text{ cm}$$

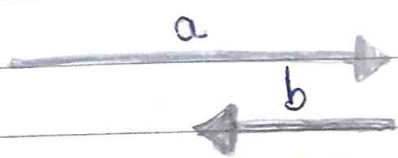
$$\tan 60 = \frac{4.5}{BC} \quad (= (\tan \theta = \frac{\text{op}}{\text{ad}} = \frac{AB}{BC}))$$

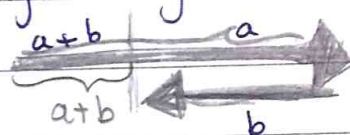
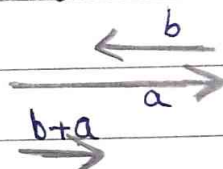
$$\cancel{BC} \cdot \cancel{(\tan 60)} = \frac{4.5}{\tan 60} = \frac{4.5}{\sqrt{3}} = 2.598 \text{ cm}$$

3. Adding vectors (one dimension)

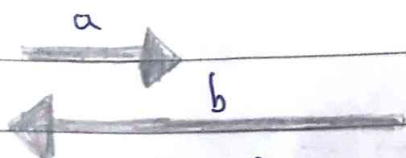
(a)  $|\vec{a}| = 6 \text{ units}$ $|\vec{b}| = 3 \text{ units}$
 ~~$|\vec{a} + \vec{b}| = 9 \text{ units}$~~
 a & b has same direction so they get added
 $a + b =$ 

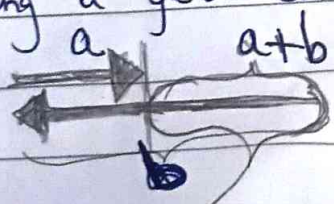
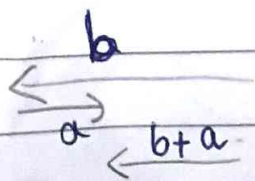
$$a + b = \text{---} \xrightarrow{a+b} = b + a$$

(b)  $|\vec{a}| = 6 \text{ units}$ $|\vec{b}| = -4 \text{ units}$
 $|\vec{a} + \vec{b}| = 6 - 4 = 2$ ~~units~~
 a & b has different direction and ^{units} $a > b$ so
 when adding b get cancelled out _{magnitude}

$$a + b =$$



$$a + b = \text{---} \xrightarrow{a+b} = b + a$$

(c)  $|\vec{a}| = +3 \text{ units}$ $|\vec{b}| = -8 \text{ units}$
 $|\vec{a} + \vec{b}| = 3 + (-8) = -5 \text{ units}$
 a & b has different direction and $b > a$, $a < b$ so
 when adding a get cancelled out ^{in magnitude}

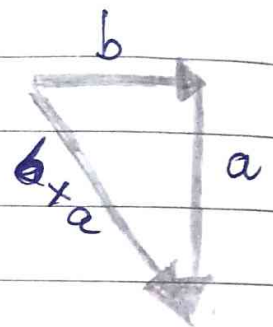
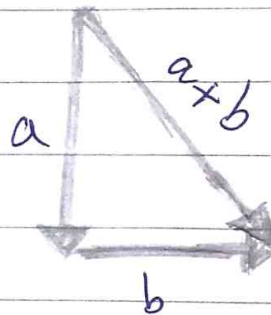
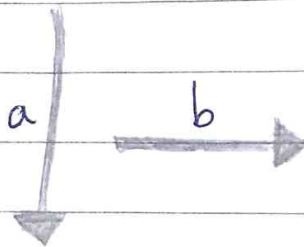
$$a + b =$$



$$a + b = \text{---} \xrightarrow{a+b} = b + a$$

from (a), (b), (c) it is clear that $\vec{a} + \vec{b} = \vec{b} + \vec{a}$
 Vector is commutative in addition
 $a + b$ give same result as $b + a$

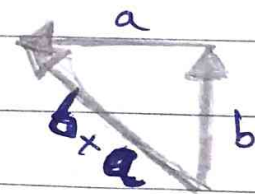
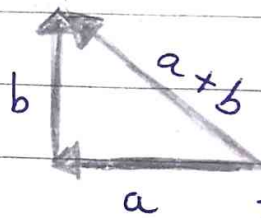
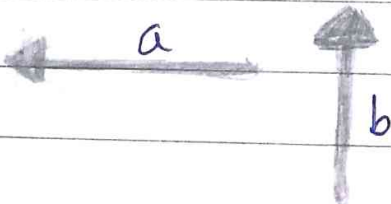
4. Adding vectors (two dimension)

(a)



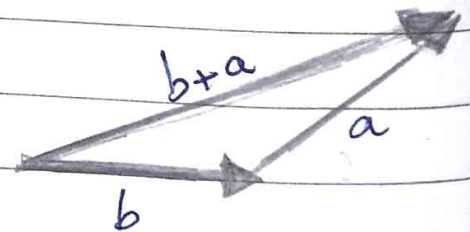
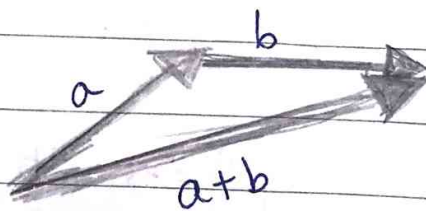
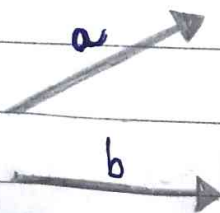
$$\vec{a} + \vec{b} = \vec{b} + \vec{a}$$

(b)



$$\vec{a} + \vec{b} = \vec{b} + \vec{a}$$

(c)

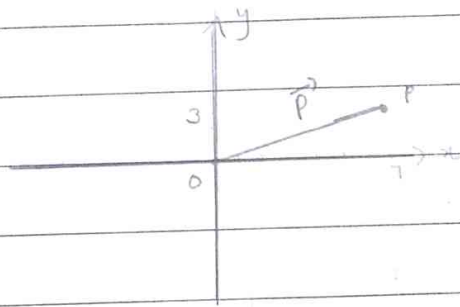


$$\vec{a} + \vec{b} = \vec{b} + \vec{a}$$

5. Column vectors

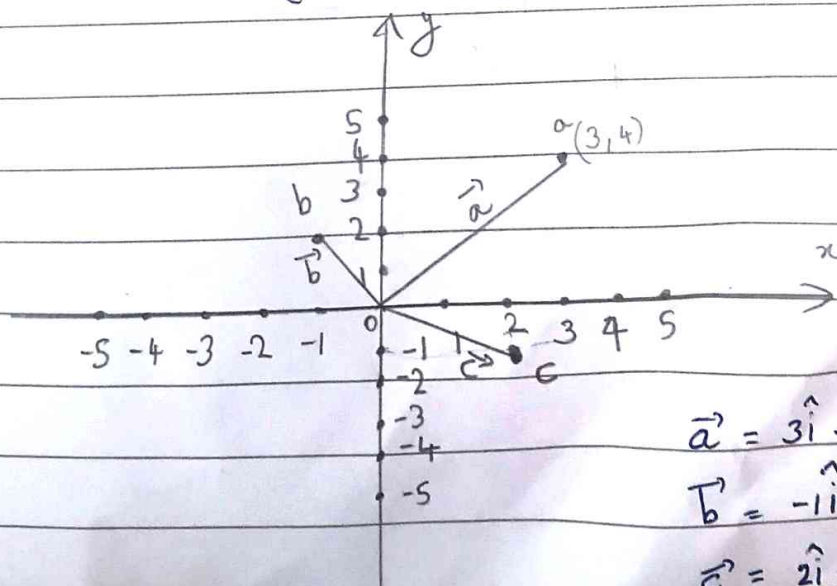
A column vector describes how a vector is to be drawn. For example, the column vector $p = \begin{pmatrix} 7 \\ 3 \end{pmatrix}$ tells us that the vector p is 7 units long in the x -direction and 3 units long in the y -direction.

The column vector ~~$p = \begin{pmatrix} 7 \\ 3 \end{pmatrix}$~~ $p = \begin{pmatrix} 7 \\ 3 \end{pmatrix}$, when drawn from the origin, will end at the point $(7, 3)$.



$$\vec{p} = 7\hat{i} + 3\hat{j}$$

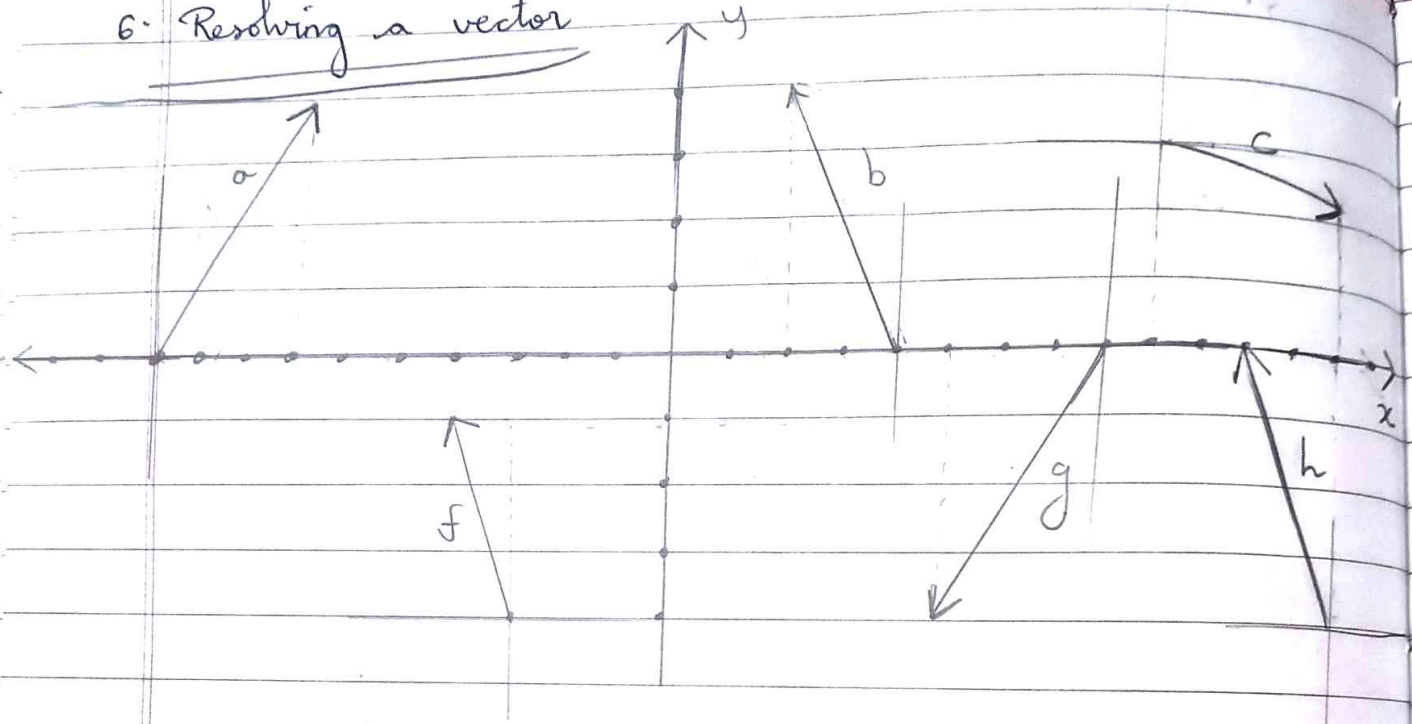
$$a = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \quad b = \begin{pmatrix} -1 \\ 2 \end{pmatrix} \quad c = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$



$$\vec{a} = 3\hat{i} + 4\hat{j}$$

$$\vec{b} = -1\hat{i} + 2\hat{j}$$

$$\vec{c} = 2\hat{i} - 1\hat{j}$$

6. Resolving a vector

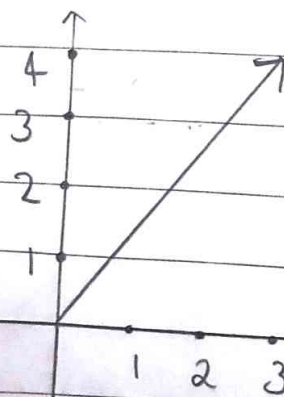
Column vectors

1)

$$\vec{a} = 3\hat{i} + 4\hat{j}$$

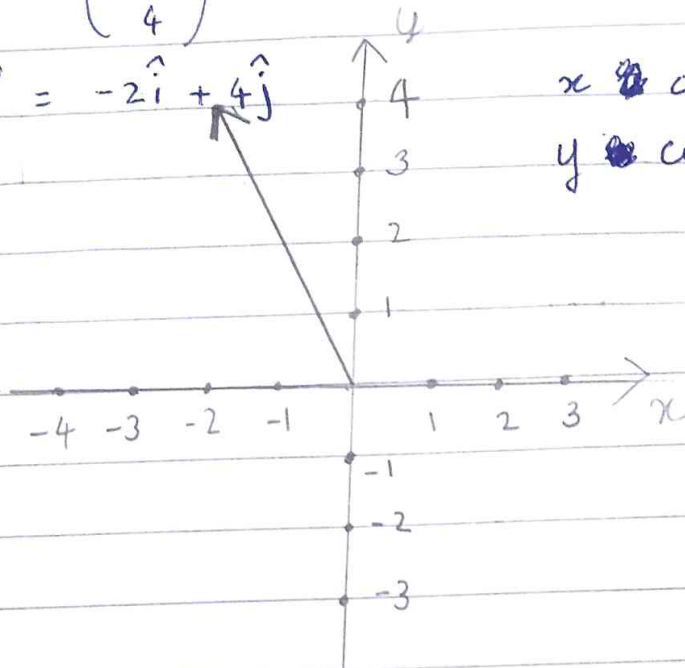
$$a = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

x component of $\vec{a}$ is 3
y component of $\vec{a}$ is 4



$$2) \quad b = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$$

$$\vec{b} = -2\hat{i} + 4\hat{j}$$

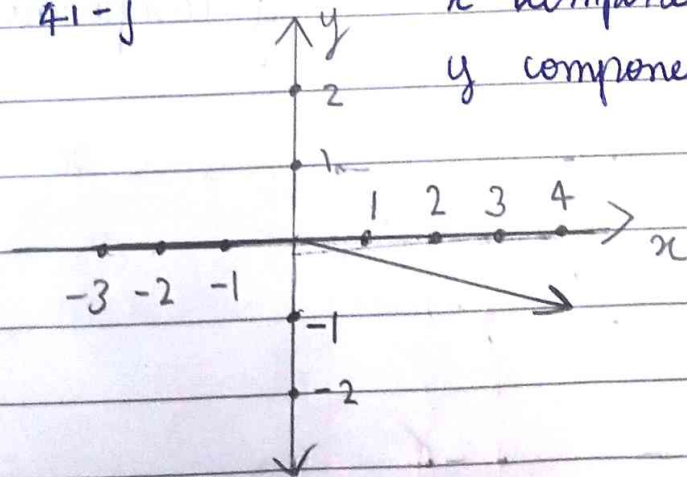


x component of $\vec{b}$ is -2

y component of $\vec{b}$ is 4

$$3) \quad c = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$$

$$\vec{c} = 4\hat{i} - \hat{j}$$

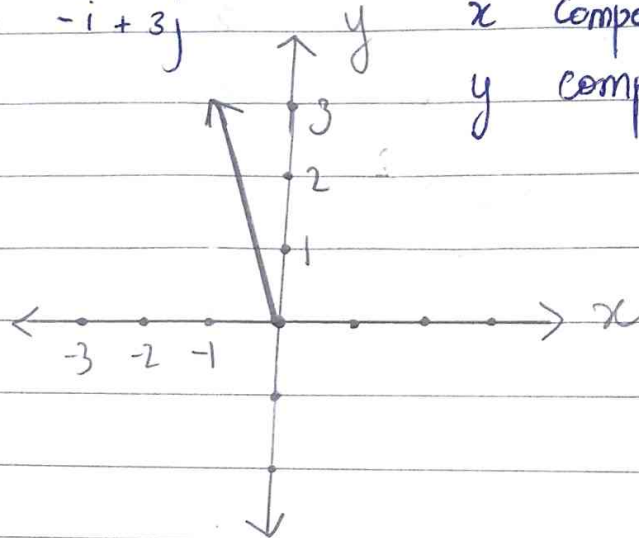


x component of $\vec{c}$ is 4

y component of $\vec{c}$ is -1

4) $\vec{f} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$

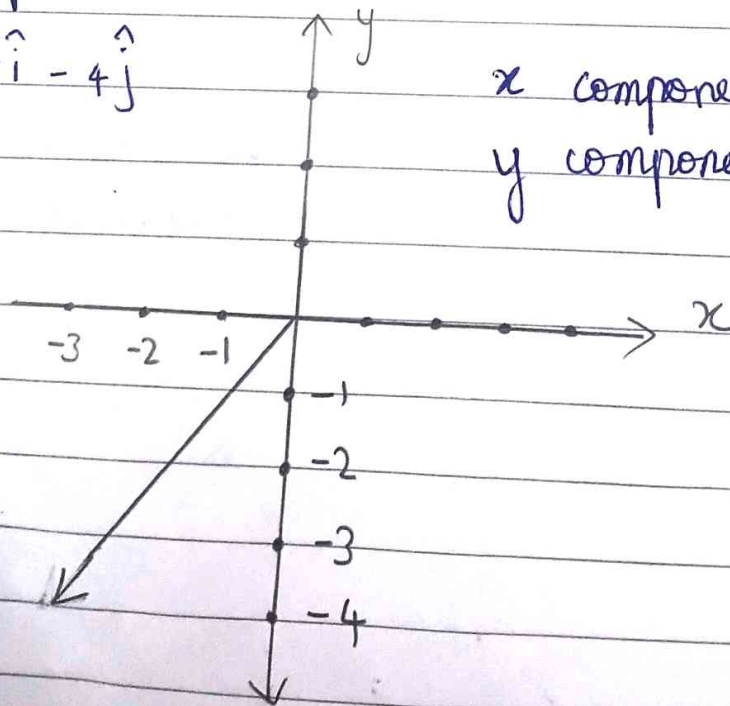
$\vec{f} = -\hat{i} + 3\hat{j}$



x component of $\vec{f}$ is -1
y component of $\vec{f}$ is 3

5) $\vec{g} = \begin{pmatrix} -3 \\ -4 \end{pmatrix}$

$\vec{g} = -3\hat{i} - 4\hat{j}$

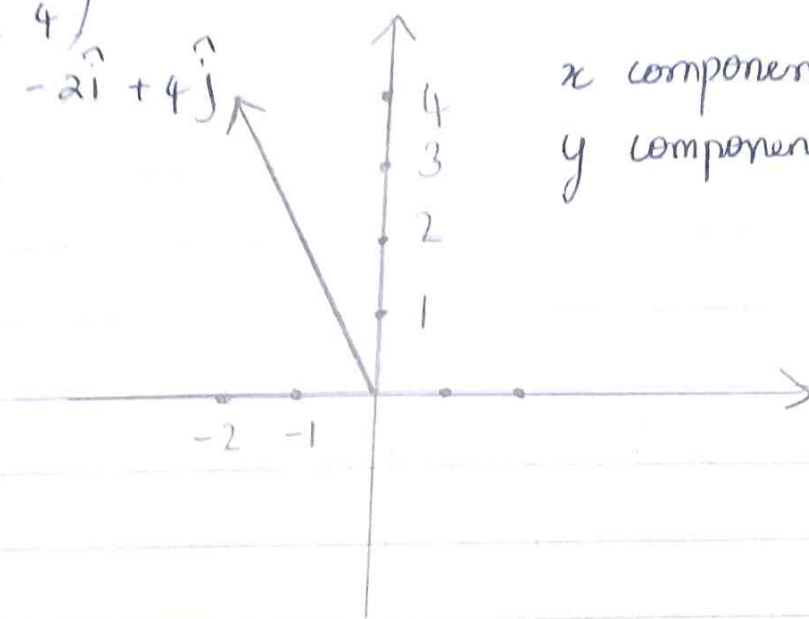


x component of $\vec{g}$ is -3
y component of $\vec{g}$ is -4

6) $h = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$

$\vec{h} = -2\hat{i} + 4\hat{j}$

x component of $\vec{h}$ is -2
y component of $\vec{h}$ is 4

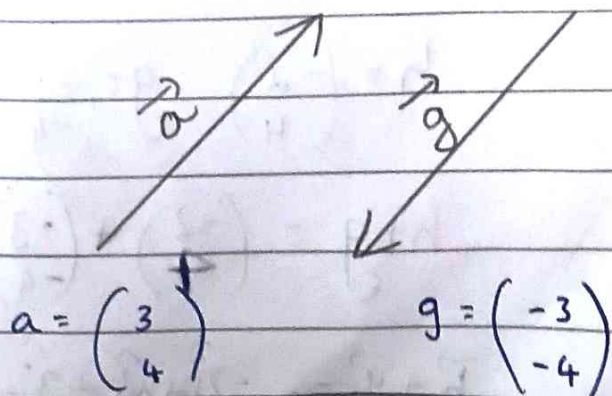


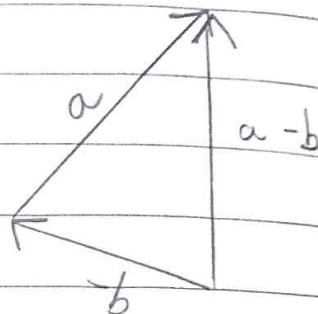
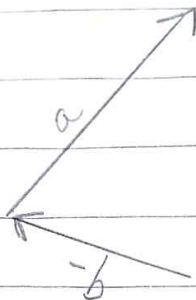
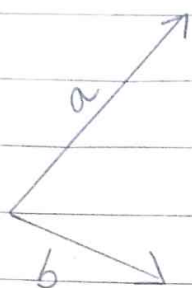
Relation between a and g

$$\vec{a} = 3\hat{i} + 4\hat{j} \quad \vec{g} = -3\hat{i} - 4\hat{j} = -(3\hat{i} + 4\hat{j}) = -\vec{a}$$

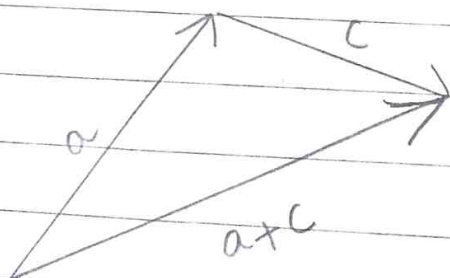
$$\therefore \boxed{\vec{g} = -\vec{a}}$$

This shows that the vectors have same magnitude but are in different directions
 $|\vec{a}| = |\vec{g}|$



7. Subtracting vectors

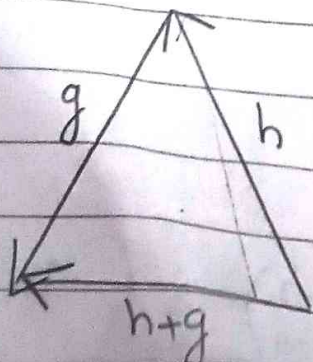
$$a - b = a + (-b)$$

8. Adding and Subtracting column vectors (from Question 6)(a) $a + c$ 

$$a = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \quad c = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$$

$$a + c = \begin{pmatrix} 3 \\ 4 \end{pmatrix} + \begin{pmatrix} 4 \\ -1 \end{pmatrix} = \begin{pmatrix} 7 \\ 3 \end{pmatrix}$$

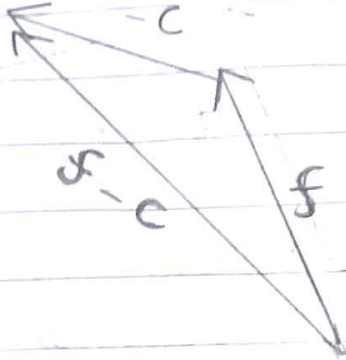
$$\vec{a} + \vec{c} = 3\hat{i} + 4\hat{j} + 4\hat{i} - \hat{j} = 7\hat{i} + 3\hat{j}$$

(b) $h + g$ 

$$h = \begin{pmatrix} -2 \\ 4 \end{pmatrix} \quad g = \begin{pmatrix} -3 \\ -4 \end{pmatrix}$$

$$h + g = \begin{pmatrix} -2 \\ 4 \end{pmatrix} + \begin{pmatrix} -3 \\ -4 \end{pmatrix} = \begin{pmatrix} -5 \\ 0 \end{pmatrix}$$

$$\vec{h} + \vec{g} = -2\hat{i} + 4\hat{j} - 3\hat{i} - 4\hat{j} = -5\hat{i} + 0\hat{j}$$

(c) $\vec{f} - \vec{c}$ 

$$\vec{f} = \begin{pmatrix} -1 \\ 3 \end{pmatrix} \quad \vec{c} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$$

$$-\vec{c} = \begin{pmatrix} -4 \\ 1 \end{pmatrix}$$

$$\vec{f} - \vec{c} = \begin{pmatrix} -1 \\ 3 \end{pmatrix} + \begin{pmatrix} -4 \\ 1 \end{pmatrix} = \begin{pmatrix} -5 \\ 4 \end{pmatrix}$$

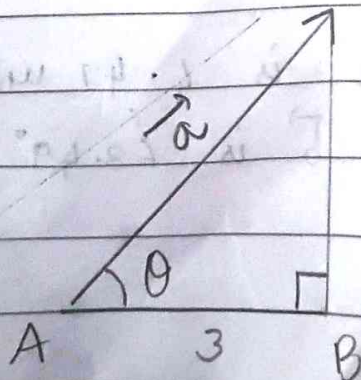
$$\begin{aligned} \vec{f} - \vec{c} &= -\hat{i} + 3\hat{j} - 4\hat{i} + \hat{j} \\ &= -5\hat{i} + 4\hat{j} \end{aligned}$$

9. Magnitude and direction of a vector

The magnitude of a vector is length of the arrow.
The direction of a vector is usually specified by taking the angle of the vector

a, b, c Q6 direction & magnitude

$$\vec{a} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \quad \vec{a} = 3\hat{i} + 4\hat{j}$$



By pythagoras theorem

$$AB^2 + BC^2 = AC^2$$

$$\begin{aligned} AC &= \sqrt{3^2 + 4^2} \\ &= \sqrt{9 + 16} = \sqrt{25} \\ &= 5 \end{aligned}$$

$$\tan \theta = \frac{\text{op}}{\text{ad}} = \frac{4}{3}$$

$$\theta = \tan^{-1}\left(\frac{4}{3}\right) = 53.13^\circ$$

magnitude of $\vec{a}$ is 5 units

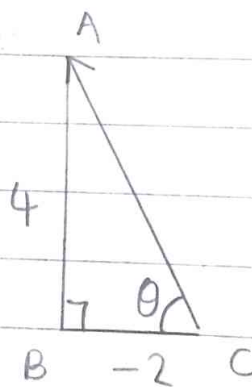
direction of $\vec{a}$ is in 53.13° w.r.t +ve x-axis

$$\vec{b} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$$

$$\vec{b} = -2\hat{i} + 4\hat{j}$$

$$\sin \theta = \frac{4}{4.47}$$

$$\theta = \sin^{-1}\left(\frac{4}{4.47}\right)$$



By Pythagoras theorem

$$AC^2 = AB^2 + BC^2$$

$$AC = \sqrt{4^2 + (-2)^2}$$

$$= \sqrt{16 + 4} = \sqrt{20}$$

$$= 2\sqrt{5} = 4.47$$

$$\sin \theta = \frac{\text{op}}{\text{hy}} = \frac{4}{4.47}$$

$$\theta = \sin^{-1}\left(\frac{4}{4.47}\right) = 63.489^\circ \approx 63.49^\circ$$

magnitude of $\vec{b}$ is 4.47 units

direction of $\vec{b}$ is 63.49° w.r.t +ve x-axis

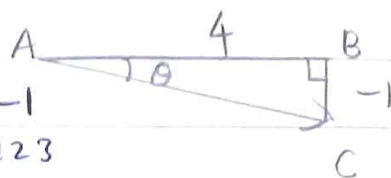
$$\vec{c} = \begin{pmatrix} 4 \\ -1 \end{pmatrix} \quad \vec{c} = 4\hat{i} - \hat{j}$$

$$\sin \theta = \frac{op}{hy} = \frac{-1}{4.123}$$

$\theta =$

$$\theta = \sin^{-1} \left(\frac{-1}{4.123} \right) = -14.036^\circ$$

$$\approx -14.04^\circ$$



Pythagoras theorem

$$AB^2 + BC^2 = AC^2$$

$$\sqrt{4^2 + (-1)^2} = AC$$

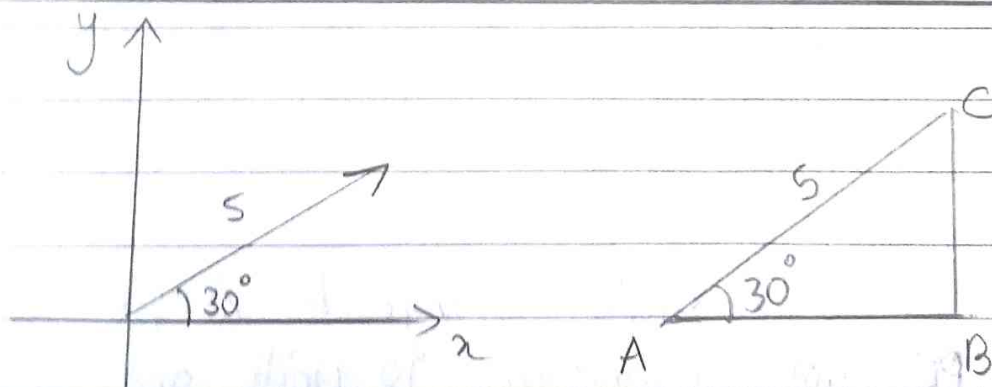
$$AC = \sqrt{16 + 1} = \sqrt{17}$$

$$= 4.123$$

magnitude of $\vec{c} = 4.123$ units

direction of $\vec{c}$ is -14.04° w.r.t +ve x-axis

10. Vectors components from magnitude and direction



let the right angle triangle be $\triangle ABC$ with $\vec{a}$ as $\vec{AC}$ with magnitude 5 units and direction 30° w.r.t +ve x-axis

$$\sin \theta = \frac{op}{hy} \Rightarrow \sin 30^\circ = \frac{BC}{5}$$

$$BC = 5 \times \frac{1}{2} = 2.5 \text{ units}$$

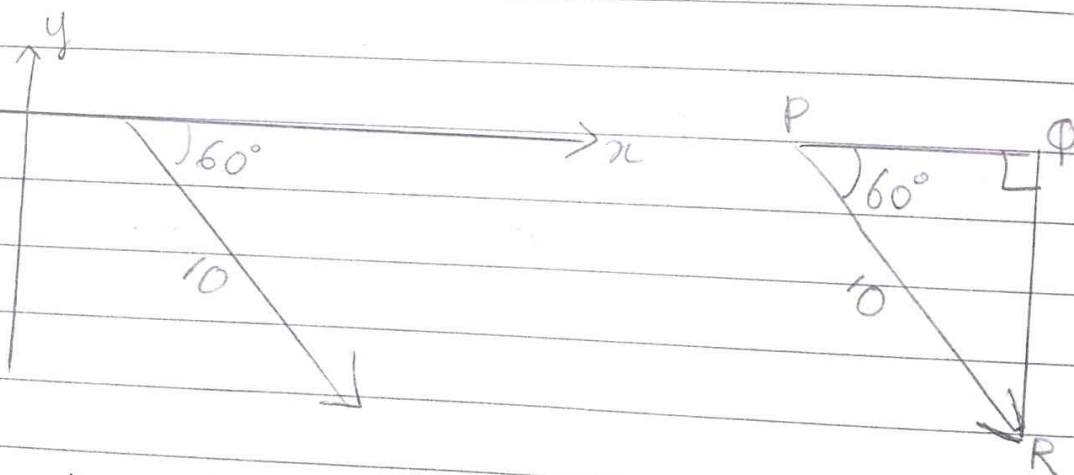
$$\cos \theta = \frac{ad}{hy} \Rightarrow \cos 30^\circ = \frac{AB}{5}$$

$$AB = 5 \times \frac{\sqrt{3}}{2} = 2.5\sqrt{3} = 4.33 \text{ units}$$

x component of $\vec{a}$ is 4.33 units

y component of $\vec{a}$ is 2.5 units

$$\vec{a} = \begin{pmatrix} 4.33 \\ 2.5 \end{pmatrix} = 4.33\hat{i} + 2.5\hat{j}$$



Let the right angled triangle be ΔPQR with $\vec{b}$ as $\vec{PR}$ with magnitude 10 units and direction 60° w.r.t +ve x-axis

$$\sin \theta = \frac{op}{hy} \Rightarrow \sin(60^\circ) = \frac{QR}{10}$$

$$QR = 10 \times \sqrt{3} = 5\sqrt{3}$$

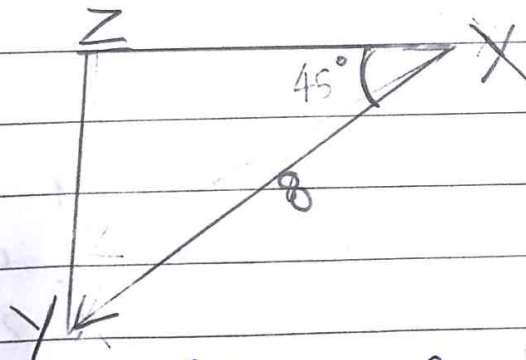
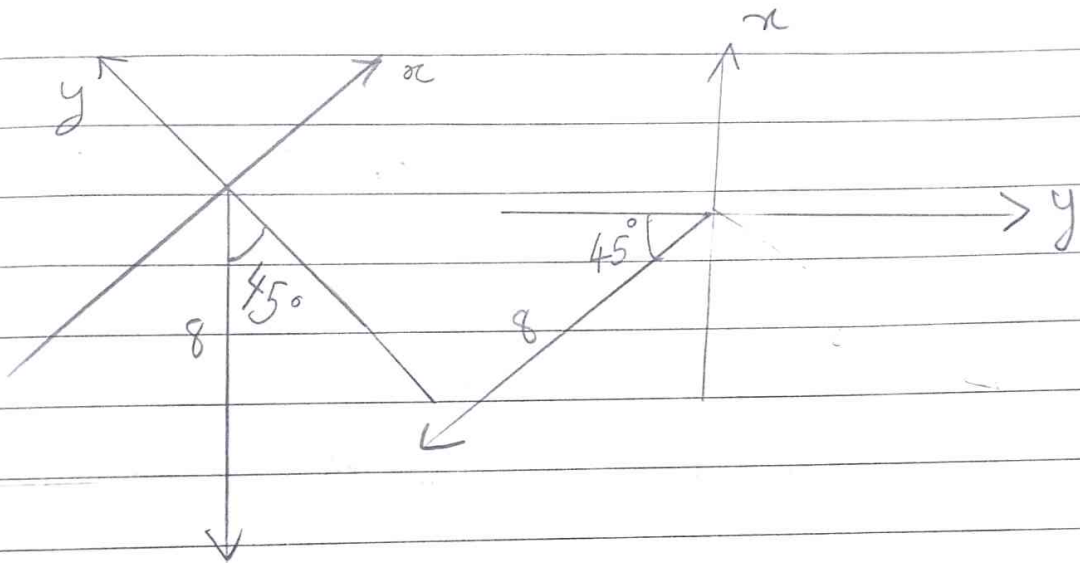
$$= 8.66^2 \text{ units } (\sin \theta) = \sin \theta$$

$$\cos \theta = \frac{ad}{hy} \Rightarrow \cos 60^\circ = \frac{PQ}{10}$$

$$PQ = 10 \times \frac{1}{2} = 5 \text{ units}$$

x component of $\vec{B}$ is 5 units
 y component of $\vec{B}$ is -8.6 units

$$\vec{B} = \begin{pmatrix} 5 \\ -8.6 \end{pmatrix} = 5\hat{i} - 8.6\hat{j}$$



Let the right angled triangle be ΔXYZ
 with $\vec{C}$ as $\vec{XY}$ with magnitude 8 units and direction 45°

$$\sin \theta = \frac{\text{op}}{\text{hy}} \Rightarrow \sin(-45^\circ) = \frac{ZY}{8}$$

$$ZY = \frac{-8}{\sqrt{2}} = -4\sqrt{2} = -5.65 \text{ units}$$

$$\cos \theta = \frac{ad}{hy} \Rightarrow \cos 45^\circ = \frac{zx}{8}$$

$$zx = \frac{8}{\sqrt{2}} = 4\sqrt{2} = 5.65 \text{ units}$$

x component of $\vec{c}$ is 5.65

y component of $\vec{c}$ is -5.65

$$\vec{c} = \begin{pmatrix} 5.65 \\ -5.65 \end{pmatrix} = 5.65\hat{i} - 5.65\hat{j}$$

11. Given : x component of velocity of a particle = 25 m/s
y component of velocity of a particle = 60 m/s

To Find : magnitude & direction of velocity vector

let $\vec{a}$ be the velocity vector with x component 25 m/s and y component 60 m/s

$$\therefore \vec{a} = \begin{pmatrix} 25 \\ 60 \end{pmatrix} = 25\hat{i} + 60\hat{j}$$

$$\text{magnitude} \Rightarrow |\vec{a}| = \sqrt{25^2 + 60^2} = \sqrt{625 + 3600} \\ = \sqrt{4225} = 65 \text{ units}$$

$$|\vec{a}| = 65 \text{ m/s}$$

$$\text{direction} \Rightarrow \theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{60}{25}\right) = \tan^{-1}\left(\frac{12}{5}\right) \\ = 67.38^\circ$$

$$\theta = 67.38^\circ = 67.4^\circ \text{ w.r.t +ve x-axis}$$

12. Given: Two forces acting at a point
10N, 20N

To Find: Resultant force of two forces acting at a point when angle b/w them are a) 60° b) 30°

a) angle b/w forces is 60°

magnitude of resultant force

$$F_1 = 10\text{ N} \quad F_2 = 20\text{ N}$$

angle b/w forces F_1 & F_2 $\theta = 60^\circ$

magnitude of resultant force R is calculated by cosine law formula

$$\begin{aligned} R &= \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos \theta} \\ &= \sqrt{10^2 + 20^2 + 2(10)(20) \cos 60} \\ &= \sqrt{100 + 400 + 400} \\ &= \sqrt{900} = 10\sqrt{9} \\ &= 26.45\text{ N} \end{aligned}$$

Direction of resultant force

angle α of resultant force w.r.t 10N is determined by using law of sines

Formula $\Rightarrow$ $\frac{F_2}{\sin \alpha} = \frac{R}{\sin \theta}$

$$\sin \alpha = \frac{F_2 \sin \theta}{R}$$

$$= \frac{20 \sin 60}{26.4}$$

$$= \frac{20\sqrt{3}}{26.4 \times 2} = \frac{10\sqrt{3}}{26.4}$$

$$= \frac{17.32}{26.4} = 0.656$$

$$\alpha = \sin^{-1} 0.656 = 40.9^\circ$$

b) angle b/w forces is 30°

magnitude of resultant force

$$F_1 = 10 \text{ N} \quad F_2 = 20 \text{ N}$$

angle b/t forces F_1 & F_2 , $\theta = 30^\circ$

magnitude of resultant force R is calculated by cosine law formula

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos \theta}$$

$$= \sqrt{10^2 + 20^2 + 2(10)(20) \cos 30}$$

$$= \sqrt{100 + 400 + 400\sqrt{3}}$$

$$= \sqrt{846.41}$$

$$= 29.09 \text{ N} \approx 29.1 \text{ N}$$

direction of resultant force

angle α of resultant force w.r.t 10N is determined by using law of sines formula

$$\Rightarrow \frac{F_2}{\sin \alpha} = \frac{R}{\sin \theta}$$

$$\sin \alpha = \frac{F_2 \sin \theta}{R}$$

$$= \frac{20 \sin 30}{29.1}$$

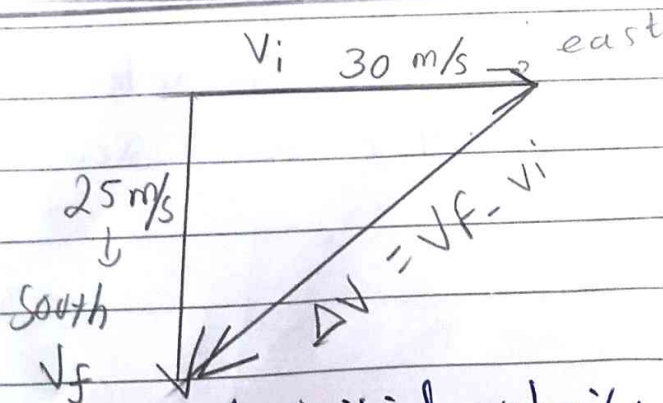
$$= \frac{20 (1)}{29.1 (2)} = \frac{10}{29.1}$$

$$= 0.343$$

$$\alpha = \sin^{-1} 0.343 = 20.05^\circ$$

$$\alpha \approx 20.1^\circ$$

13.



V_i initial velocity = 30 m/s due east
 V_f final velocity = 25 m/s due south

$$\Delta \vec{V} \text{ change in velocity} = \vec{V_f} - \vec{V_i}$$

$$|\Delta \vec{v}| = \sqrt{(\vec{v}_f)^2 + (\vec{v}_i)^2}$$

$$\vec{v}_i = -30 \hat{i}$$

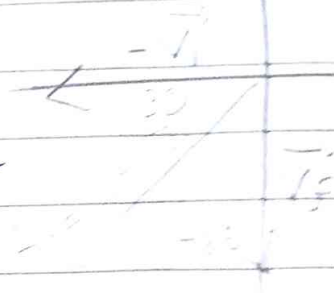
$$\vec{v}_f = -25 \hat{j}$$

$$|\Delta \vec{v}| = \sqrt{(-25)^2 + (-30)^2}$$

$$= \sqrt{625 + 900}$$

$$= \sqrt{1525}$$

$$= 39.05 \text{ m/s}$$



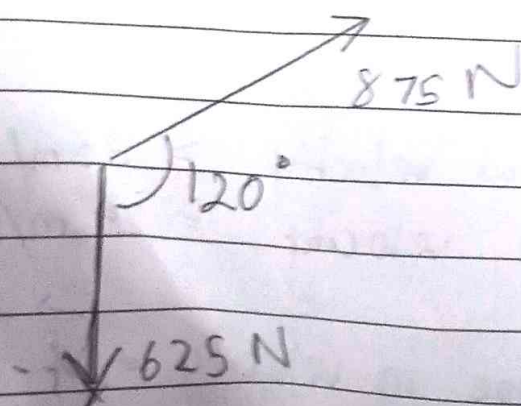
$$\tan \theta = \frac{-25}{-30} = \frac{5}{6}$$

$$\theta = \tan^{-1}\left(\frac{5}{6}\right) = 39.8^\circ$$

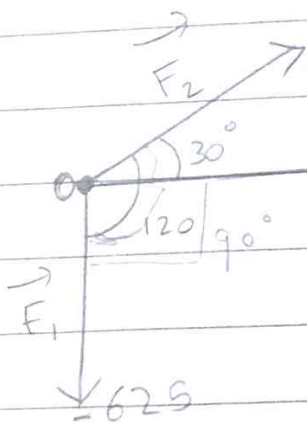
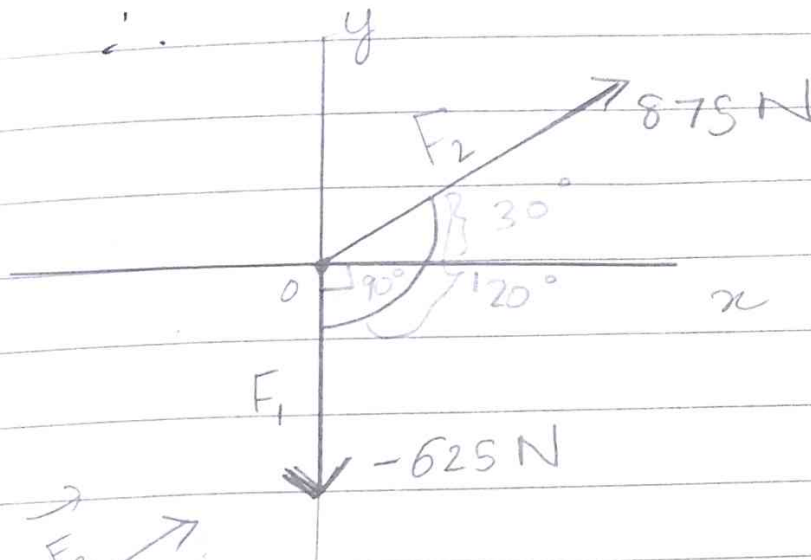
$\therefore$ magnitude of change in velocity of car
 $\Rightarrow 39.05 \text{ m/s}$

direction of change in velocity of car
 $\Rightarrow 39.8^\circ$ due south-west

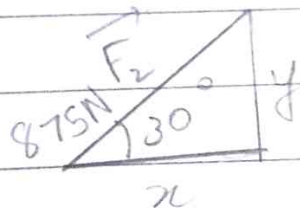
14)



given : 625 N vector is along the -y-axis
 & +x-axis is $\perp$ to it toward the right



Components of $\vec{F}_1 = \begin{pmatrix} 0 \\ -625 \end{pmatrix}$



$$\vec{F}_2 = \begin{pmatrix} 757.8 \\ 437.5 \end{pmatrix}$$

$$\sin 30^\circ = \frac{y}{875} \quad y = 875 \cdot \frac{1}{2} = 437.5$$

$$\cos 30^\circ = \frac{x}{875} \quad x = 875 \cdot \frac{\sqrt{3}}{2} = 757.77 = 757.8$$

Resultant Force Vector $\vec{R} = \vec{F}_1 + \vec{F}_2$

$$= \begin{pmatrix} 0 \\ -625 \end{pmatrix} + \begin{pmatrix} 757.8 \\ 437.5 \end{pmatrix}$$

$$= \begin{pmatrix} 757.8 \\ -187.5 \end{pmatrix}$$

magnitude of resultant force

$$|\vec{R}| = \sqrt{(757.8)^2 + (-187.5)^2}$$

$$= \sqrt{609417.09}$$

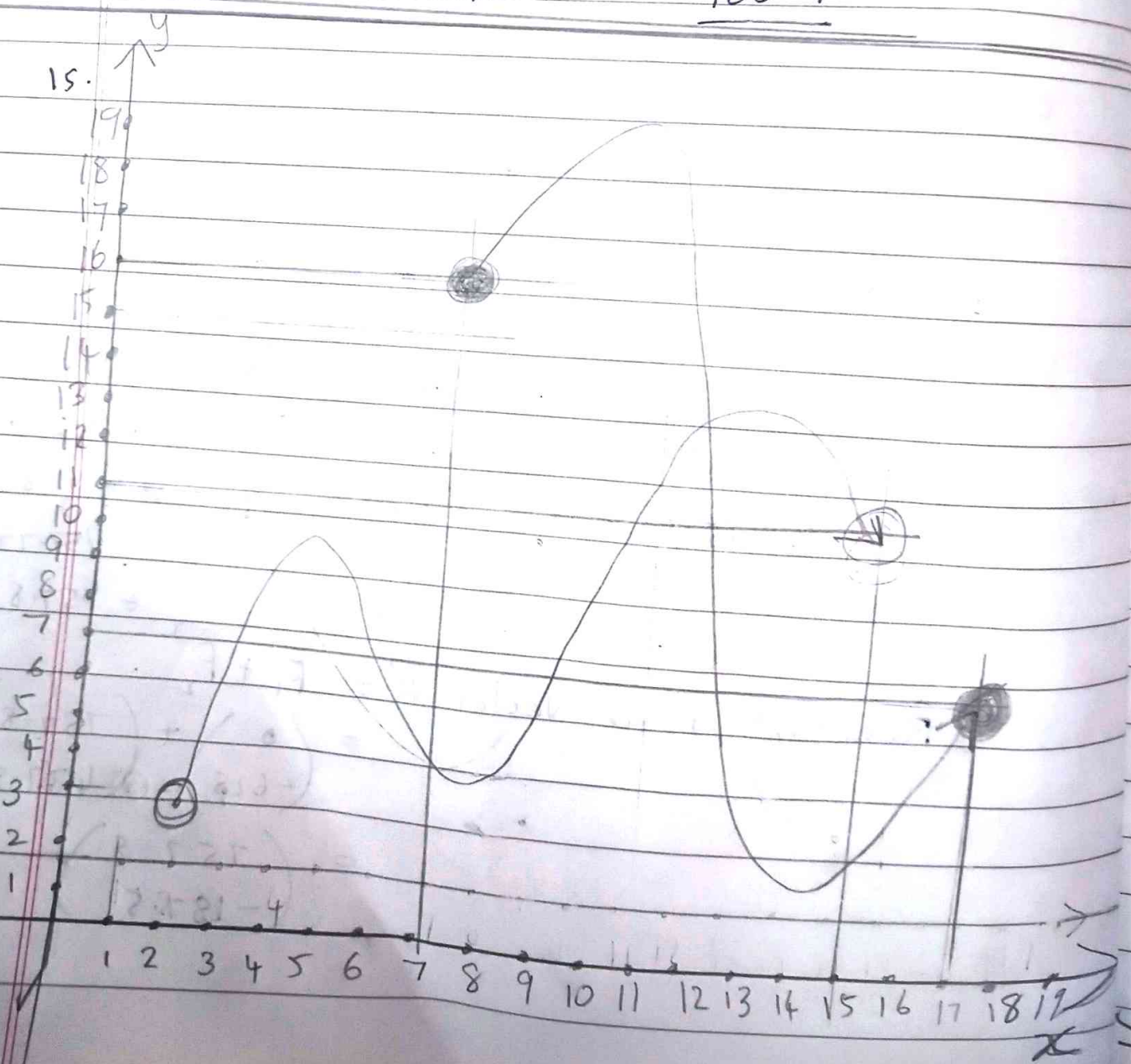
$$= 780.65 \approx 781 \text{ N}$$

direction of resultant force

$$\tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-187.5}{757.8}\right) = -13.89$$

$$\approx -13.9$$

$$180^\circ - 13.9^\circ = 166.1^\circ$$



let starting point and end point of white piece be W_i & W_f

let starting point & end point of Black piece be B_i & B_f

starting position of White piece = $(2, 3)$

ending position of White piece = $(15, 11)$

starting position of Black piece = $(7, 16)$

ending position of Black piece = $(17, 7)$

$$\vec{W}_i = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad \vec{W}_f = \begin{pmatrix} 15 \\ 11 \end{pmatrix} \quad \vec{B}_i = \begin{pmatrix} 7 \\ 16 \end{pmatrix}$$

$$\vec{B}_f = \begin{pmatrix} 17 \\ 7 \end{pmatrix}$$

$$\Delta \vec{W} = \vec{W}_f - \vec{W}_i = \begin{pmatrix} 15 \\ 11 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 13 \\ 8 \end{pmatrix}$$

$$\Delta \vec{B} = \vec{B}_f - \vec{B}_i = \begin{pmatrix} 17 \\ 7 \end{pmatrix} - \begin{pmatrix} 7 \\ 16 \end{pmatrix} = \begin{pmatrix} 10 \\ -9 \end{pmatrix}$$

Change of displacement vector

$$\Delta \vec{W} - \Delta \vec{B} = \begin{pmatrix} 13 \\ 8 \end{pmatrix} - \begin{pmatrix} 10 \\ -9 \end{pmatrix} = \begin{pmatrix} 3 \\ 17 \end{pmatrix}$$

No the actual path moved by the piece do not matter because displacement only accounts for final & initial values